Layer 0: Wound Biology

Protease Cascade ODE: MMP-9 and HNE in Wound Fluid

Eton iGEM 2026, Dry Lab | Layer 0 of 17-Layer Biosensor Pipeline | `wound_biology.py`

Layer 0 predicts the **free protease concentrations in wound fluid at the time of swab sampling**. Its two key outputs, $[\text{MMP-9}]_{\text{free}}(t)$ and $[\text{HNE}]_{\text{free}}(t)$, are in-situ wound fluid concentrations. Layer 1 (sample collection) applies a swab dilution factor D_f to convert these to the concentrations entering the electrochemical device; the sensor detection range (17–200 ng/mL MMP-9, equivalent to 0.185–2.17 nM) is reached only after that dilution.

The model tracks nine coupled species through a system of ordinary differential equations (Section IV), solved numerically with a stiff implicit integrator. Every parameter value must be grounded in a paper physically present in `papers/layer_00/`; values that are estimates rather than text-confirmed are explicitly labelled throughout.

I. Wound Fluid Composition

Biological rationale

Chronic wounds are characterised by a prolonged inflammatory phase sustained by persistent neutrophil and macrophage infiltration [1]. This drives elevated secretion of two serine and metalloprotease families that are the targets of this device. MMP-9 (gelatinase B, a zinc-dependent matrix metalloprotease) is secreted as an inactive zymogen (pro-MMP-9); activation requires proteolytic removal of the prodomain. HNE (human neutrophil elastase) is released already active from azurophil granules and is regulated primarily by α_1-antitrypsin (A1AT) and secondarily by secretory leukocyte protease inhibitor (SLPI).

Unit conversion

All species concentrations are held internally in mol/L. The conversion from the ng/mL units reported in ELISA literature is:

$$C\left[\frac{\text{mol}}{\text{L}}\right] = \frac{c\left[\frac{\text{ng}}{\text{mL}}\right] \times 10^{-6}\left[\frac{\text{g/L}}{\text{ng/mL}}\right]}{M_W\left[\frac{\text{g}}{\text{mol}}\right]} \tag{I.1}$$

For μg/mL values the prefactor is 10^{-3} instead of 10^{-6}. Applying Eq. (I.1) to active MMP-9 in chronic wound fluid (4800 ng/mL, $M_W = 92{,}000$ g/mol):

$$C_{\text{MMP-9}} = \frac{4800 \times 10^{-6}}{92{,}000} = 5.22 \times 10^{-8}\ \text{mol/L} = 52\ \text{nM}$$

Literature values

Ladwig et al. (2002) [2] used quantitative ELISA and gelatin zymography on wound fluid from 56 patients with chronic pressure ulcers. All values below are from paper text (p. 31):

- Pro-MMP-9: $8{,}100 \pm 3{,}000$ ng/mL; active MMP-9: $4{,}800 \pm 1{,}300$ ng/mL.
- Total MMP-9 = 12,900 ng/mL; pro-form fraction = 63%.
- MMP-9:TIMP-1 ratio inversely correlated with healing (text pp. 33–34): higher active MMP-9 and lower TIMP-1 in non-healing wounds.

Trengove et al. (1999) [1] used ELISA in venous leg ulcer (VLU) wound fluid. Values from text (p. 448):

- TIMP-1 in chronic VLU: median 0.8 μg/mL = 800 ng/mL ($n = 5$).
- HNE in non-healing VLU: median 1.4 μg/mL $\approx$ 47 nM (range $<$ 1–11.9 μg/mL; elevated in 6 of 13 chronic samples).
- MMP activity significantly lower in healing wounds ($p < 0.01$, text p. 449).

Rao et al. (1995) [7] examined A1AT integrity by western blot and elastase-inhibitory capacity (text pp. 572–575):

- A1AT intact and functional in all healing and acute wound samples.
- A1AT degraded and non-functional in 9 of 10 chronic wound samples.
- No quantitative ng/mL wound-fluid A1AT concentration is given in Rao 1995. The healing-state estimate $[A_0] \approx 1.92$ μM (serum $\approx$ 2 mg/mL at 10-fold dilution into wound fluid) is used; Section III shows the qualitative conclusion is insensitive to this estimate.

Nwomeh et al. (1999) [8] reported that TIMP-1 was "very low or not detected" in chronic ulcer fluid (text p. 193), consistent with Trengove 1999 [1] and confirming TIMP-1 suppression as a hallmark of the chronic wound environment. Values are expressed per μg protein and cannot be converted to ng/mL without the protein content data, so Trengove 1999 is used for quantitative TIMP-1.

Initial conditions

Table 1 gives the initial conditions for both wound states.

Species	Chronic wound	Healing wound	Confirmed?	Source
Total MMP-9	12,900 ng/mL (140 nM)	5,400 ng/mL (59 nM)	Chronic yes / Healing fig.	[2]
Pro-MMP-9 (63%)	8,100 ng/mL (88 nM)	3,400 ng/mL (37 nM)	Chronic text / Healing est.	[2]
Active MMP-9 (37%)	4,800 ng/mL (52 nM)	2,000 ng/mL (22 nM)	Chronic text / Healing fig.	[2]
TIMP-1	800 ng/mL (28 nM)	1,104 ng/mL (39 nM)	Chronic text / Healing none	[1]
HNE	47 nM (1.4 μg/mL), both states		Text	[1]
A1AT (functional)	≈ 0 (degraded)	1.92 μM (est.)	Qualitative	[7]
SLPI	≈ 200 ng/mL (≈ 17 nM)		Not confirmed	est.

Table 1: Initial conditions for the Layer 0 ODE. Healing MMP-9 is read from Ladwig 2002 Figure 8 (approximate, not stated in text). Healing TIMP-1 has no paper source. A1AT wound-state status is qualitative from Rao 1995 [7] with an estimated concentration.

II. MMP-9 Inhibition by TIMP-1

TIMP-1 binds reversibly to the MMP-9 active site in non-covalent slow-tight inhibition. Denoting free MMP-9 as M, free TIMP-1 as T, and their complex as C_1, the rate equations are:

$$\frac{d[M]}{dt} = -k_{\text{on}}^{T}\,[M][T] + k_{\text{off}}^{T}\,[C_1] \tag{II.1}$$

$$\frac{d[T]}{dt} = -k_{\text{on}}^{T}\,[M][T] + k_{\text{off}}^{T}\,[C_1] \tag{II.2}$$

$$\frac{d[C_1]}{dt} = +k_{\text{on}}^{T}\,[M][T] - k_{\text{off}}^{T}\,[C_1] \tag{II.3}$$

Olson et al. (1997) [4] reported kinetic constants from solution-phase enzyme inhibition assays (Table IV, 82 kDa active MMP-9):

$$k_{\text{on}}^{T} = 2.48 \pm 0.47 \times 10^{5}\ \text{M}^{-1}\text{s}^{-1}$$
$$k_{\text{off}}^{T} = 2.1 \pm 1.1 \times 10^{-3}\ \text{s}^{-1}$$

The equilibrium inhibition constant K_i is stated directly in Olson 1997 Table IV as $K_i = 8.5 \pm 2.9$ nM; the ratio confirms this:

$$K_i = \frac{k_{\text{off}}^{T}}{k_{\text{on}}^{T}} = \frac{2.1 \times 10^{-3}}{2.48 \times 10^{5}} = 8.47\ \text{nM} \approx 8.5\ \text{nM} \tag{II.4}$$

Enzyme inhibition values are used in preference to the SPR values in Olson 1997 Tables I and II because solution-phase kinetics are more representative of wound-fluid conditions; k_{on}^{T} from enzyme inhibition is 2.5–7× higher than from SPR (Olson 1997, p. 29981).

Equilibrium free MMP-9

The TIMP-1/MMP-9 equilibration timescale is:

$$\tau = \frac{1}{k_{\text{on}}^{T}\,[T_0]} = \frac{1}{2.48 \times 10^{5} \times 28 \times 10^{-9}} \approx 144\ \text{s} \approx 2.4\ \text{min}$$

This is short relative to the 30-min measurement window, so the system rapidly reaches an approximate binding equilibrium. Setting $d[C_1]/dt = 0$ in Eq. (II.3) gives:

$$K_i = \frac{[M][T]}{[C_1]} \tag{II.5}$$

Conservation of each species pool requires:

$$[M_0] = [M] + [C_1] \tag{II.6}$$
$$[T_0] = [T] + [C_1] \tag{II.7}$$

From Eq. (II.6): $[C_1] = [M_0] - [M]$. Substituting into Eq. (II.7): $[T] = [T_0] - [M_0] + [M]$. Substituting both into Eq. (II.5) and writing $x = [M]_{\text{free}}$:

$$K_i\,(M_0 - x) = x\,(T_0 - M_0 + x)$$

Expanding and collecting terms:

$$x^2 + x\,(T_0 - M_0 + K_i) - K_i\,M_0 = 0 \tag{II.8}$$

Eq. (II.8) is a quadratic in x. Taking the positive root (the only physically meaningful solution, since $[M]_{\text{free}} \geq 0$):

$$[M]_{\text{free}} = \frac{-(T_0 - M_0 + K_i) + \sqrt{(T_0 - M_0 + K_i)^2 + 4\,K_i\,M_0}}{2} \tag{II.9}$$

Chronic wound ($M_0 = 52$ nM, $T_0 = 28$ nM, $K_i = 8.5$ nM):

$$b = T_0 - M_0 + K_i = 28 - 52 + 8.5 = -15.5 \text{ nM}$$
$$[M]_{\text{free}} = \frac{15.5 + \sqrt{(-15.5)^2 + 4 \times 8.5 \times 52}}{2}$$
$$= \frac{15.5 + \sqrt{240.3 + 1768}}{2} = \frac{15.5 + 44.8}{2} \approx \mathbf{30.2} \text{ nM}$$

Healing wound ($M_0 = 22$ nM, $T_0 = 39$ nM, $K_i = 8.5$ nM):

$$b = T_0 - M_0 + K_i = 39 - 22 + 8.5 = +25.5 \text{ nM}$$
$$[M]_{\text{free}} = \frac{-25.5 + \sqrt{(25.5)^2 + 4 \times 8.5 \times 22}}{2}$$
$$= \frac{-25.5 + \sqrt{650.3 + 748}}{2} = \frac{-25.5 + 37.4}{2} \approx \mathbf{5.9} \text{ nM}$$

These analytical equilibrium values agree closely with the full ODE result at $t = 30$ min (31.9 nM chronic, 6.2 nM healing; Table 2). The small difference reflects ongoing pro-MMP-9 activation driving the system slightly away from the pure TIMP-1 equilibrium.

Sensitivity to K_i

Changing K_i from 8.5 nM to 1 nM in Eq. (II.9) shifts the chronic result from 30.2 nM to 26.5 nM (a 12% decrease). The healing result is more sensitive: at $K_i = 1$ nM it falls from 5.9 nM to 1.4 nM, because the TIMP-1 excess (17 nM) is comparable to K_i. The reported uncertainty (±2.9 nM) is within this regime; the healing estimate should be treated as order-of-magnitude.

The following assumptions underlie this section:

1. The equilibrium approximation is valid because $\tau \approx 2.4$ min is short relative to the 30-min read window and the pro-MMP-9 activation timescale.
2. Solution-phase enzyme inhibition constants (Olson 1997 [4] Table IV) are used in preference to SPR values, as the wound fluid environment is closer to solution phase.
3. Pro-MMP-9 activation (Section IV) is negligible with current placeholder kinetics; the active MMP-9 initial condition is approximately constant over the measurement window.

III. HNE Inhibition

A1AT/HNE near-irreversible binding

A1AT inhibits HNE via serpin mechanism: a covalent acyl-enzyme intermediate forms and the complex is effectively irreversible ($k_{\text{off}}^{A} \approx 0$). The rate of free HNE depletion is:

$$\frac{d[H]}{dt} = -k_{\text{on}}^{A}\,[A][H] \tag{III.1}$$

In healing wounds where $[A_0] \gg [H_0]$ (see below), the free A1AT concentration remains approximately constant throughout the reaction. This pseudo-first-order approximation reduces Eq. (III.1) to:

$$\frac{d[H]}{dt} \approx -k^{*}\,[H], \qquad k^{*} = k_{\text{on}}^{A}\,[A_0] \tag{III.2}$$

Eq. (III.2) is a first-order linear ODE with exponential solution:

$$[H](t) = [H_0]\,\exp\!\big(-k_{\text{on}}^{A}\,[A_0]\,t\big) \tag{III.3}$$

The half-life, at which $[H] = [H_0]/2$, follows from setting $\exp(-k^{*}t_{1/2}) = 1/2$:

$$t_{1/2} = \frac{\ln 2}{k_{\text{on}}^{A} \cdot [A_0]} \tag{III.4}$$

Beatty, Bieth & Travis (1980) [5] confirmed $k_{\text{on}}^{A} = 6.5 \pm 4.0 \times 10^{7}$ $\text{M}^{-1}\text{s}^{-1}$ from direct kinetic measurement (Table 1, human leukocyte elastase vs. native α_1-PI). With the healing-wound estimate $[A_0] = 1.92$ μM:

$$t_{1/2} = \frac{0.693}{6.5 \times 10^{7} \times 1.92 \times 10^{-6}} = \frac{0.693}{124.8} \approx 5.5 \text{ ms} \tag{III.5}$$

Sensitivity to $[A_0]$. At $[A_0]$ ten-fold lower (0.192 μM):

$$t_{1/2} = \frac{0.693}{6.5 \times 10^{7} \times 0.192 \times 10^{-6}} \approx 55 \text{ ms} \tag{III.6}$$

Even at this lower bound, $t_{1/2} = 55$ ms remains three orders of magnitude below the 5-min practical minimum measurement window. The healing-wound conclusion, $[\text{HNE}]_{\text{free}} \approx 0$, is therefore insensitive to the [A1AT] estimate over a 10-fold range.

Validity check. In healing wounds $[A_0] = 1.92$ $\mu\text{M} \gg [H_0] = 47$ nM, confirming the pseudo-first-order approximation. After complete HNE absorption: $[A]_{\text{free}} \approx 1.92$ $\mu\text{M} - 47$ nM ≈ 1.87 μM (confirmed by ODE, Table 2).

A1AT wound-state dependence

Rao et al. (1995) [7] showed that functional A1AT cannot be assumed present across wound types:

- **Healing and acute wounds**: A1AT intact and functional in all samples. Eq. (III.5) applies; $[\text{HNE}]_{\text{free}} = 0$ within milliseconds.
- **Chronic wounds**: A1AT degraded and non-functional in 9 of 10 samples. With functional $[A_0] \approx 0$, Eq. (III.1) gives $d[H]/dt \approx 0$: HNE persists in wound fluid at near its initial concentration (moderated only by SLPI).

This A1AT wound-state switch provides a second discriminating axis: healing wounds have $[\text{HNE}]_{\text{free}} \approx 0$ while chronic wounds (where A1AT is degraded) retain elevated free HNE. However, HNE elevation itself was detected in only 6 of 13 chronic VLU samples in Trengove 1999 [1], meaning ~50% of chronic cases are HNE-low regardless of A1AT status. HNE is therefore not a reliable standalone discriminator (see also Section V).

SLPI secondary inhibition in chronic wounds

In chronic wounds where A1AT is non-functional, SLPI provides the residual HNE inhibition. Ashcroft et al. (2000) [6] confirmed SLPI's role in wound healing using knockout mice but did not report a quantitative wound-fluid SLPI concentration. The model uses placeholder kinetics ($k_{\text{on}}^S = 10^5$ M^{-1}s^{-1}, $k_{\text{off}}^S = 10^{-4}$ s^{-1}, $[S_0] \approx 200$ ng/mL ≈ 17 nM); with these values the ODE gives $[\text{HNE}]_{\text{free}} \approx 30$ nM at $t = 30$ min in chronic wounds. This prediction is sensitive to the placeholder SLPI kinetics.

The following assumptions underlie this section:

1. Pseudo-first-order kinetics are valid for healing wounds because $[A_0] = 1.92\ \mu\text{M} \gg [H_0] = 47$ nM.
2. $[A_0] = 1.92\ \mu$M is a serum/10 estimate; the half-life conclusion is insensitive over a 10-fold range (Eq. (III.6)).
3. k_{on}^A was measured at 25°C, pH 8.0 in Beatty 1980 [5]; applied here at wound pH 7.4 as the best available value.
4. SLPI concentration and kinetics are unconfirmed placeholders; the chronic HNE prediction is not considered quantitatively reliable.

IV. Full ODE System

Species and notation

Nine coupled species track the wound fluid protease cascade. All concentrations in mol/L:

P	pro-MMP-9	C_1	MMP-9:TIMP-1 complex
M	free (active) MMP-9	H	free HNE
T	free TIMP-1	A	free A1AT
C_2	HNE:A1AT complex	S	free SLPI
C_3	HNE:SLPI complex		

Governing equations

Pro-MMP-9 is activated by MMP-3 and plasmin in excess, so the activation is pseudo-first-order in P with effective rate k^{**}:

$$\dot{P} = -k^{**}\,P \tag{IV.1}$$
$$\dot{M} = +k^{**}\,P - k_{\text{on}}^T MT + k_{\text{off}}^T C_1 \tag{IV.2}$$
$$\dot{T} = -k_{\text{on}}^T MT + k_{\text{off}}^T C_1 \tag{IV.3}$$
$$\dot{C}_1 = +k_{\text{on}}^T MT - k_{\text{off}}^T C_1 \tag{IV.4}$$
$$\dot{H} = -k_{\text{on}}^A HA - k_{\text{on}}^S HS + k_{\text{off}}^S C_3 \tag{IV.5}$$
$$\dot{A} = -k_{\text{on}}^A HA \tag{IV.6}$$
$$\dot{C}_2 = +k_{\text{on}}^A HA \tag{IV.7}$$
$$\dot{S} = -k_{\text{on}}^S HS + k_{\text{off}}^S C_3 \tag{IV.8}$$
$$\dot{C}_3 = +k_{\text{on}}^S HS - k_{\text{off}}^S C_3 \tag{IV.9}$$

where $k^{**} = (k_{\text{cat}}/K_m)_{\text{MMP-3}}\,[\text{MMP-3}] + (k_{\text{cat}}/K_m)_{\text{plasmin}}\,[\text{plasmin}]$ is the total first-order activation rate; k_{on}^T, k_{off}^T from Olson 1997 [4]; k_{on}^A from Beatty 1980 [5]; k_{on}^S, k_{off}^S are placeholders.

Mass conservation

Adding the appropriate equations confirms that each species pool is separately conserved:

$$\tfrac{d}{dt}(P + M + C_1) = 0 \qquad \text{(total MMP-9)}$$
$$\tfrac{d}{dt}(T + C_1) = 0 \qquad \text{(total TIMP-1)}$$
$$\tfrac{d}{dt}(H + C_2 + C_3) = 0 \qquad \text{(total HNE)}$$
$$\tfrac{d}{dt}(A + C_2) = 0 \qquad \text{(total A1AT)}$$
$$\tfrac{d}{dt}(S + C_3) = 0 \qquad \text{(total SLPI)}$$

These identities provide a numerical check: deviations indicate integration error. The Radau solver maintains conservation to within rtol $= 10^{-10}$, atol $= 10^{-15}$ tolerances.

Stiff integration

The system spans seven orders of magnitude in characteristic timescale:

- A1AT/HNE reaction: $t_{1/2} \approx 5.5$ ms (fastest)
- TIMP-1/MMP-9 equilibration: $\tau \approx 2.4$ min (intermediate)
- Pro-MMP-9 activation: $\sim$120 min (slowest; negligible with placeholders)

The timescale ratio $\sim 10^7$ makes the system stiff: explicit solvers would require $\Delta t \lesssim 1$ ms, making a 120-min integration prohibitively expensive. The Radau implicit solver (`scipy.integrate.solve_ivp`, method `Radau`) is used throughout.

V. Model Output and Wound-State Discrimination

Figures 1 and 2 show the ODE trajectories.

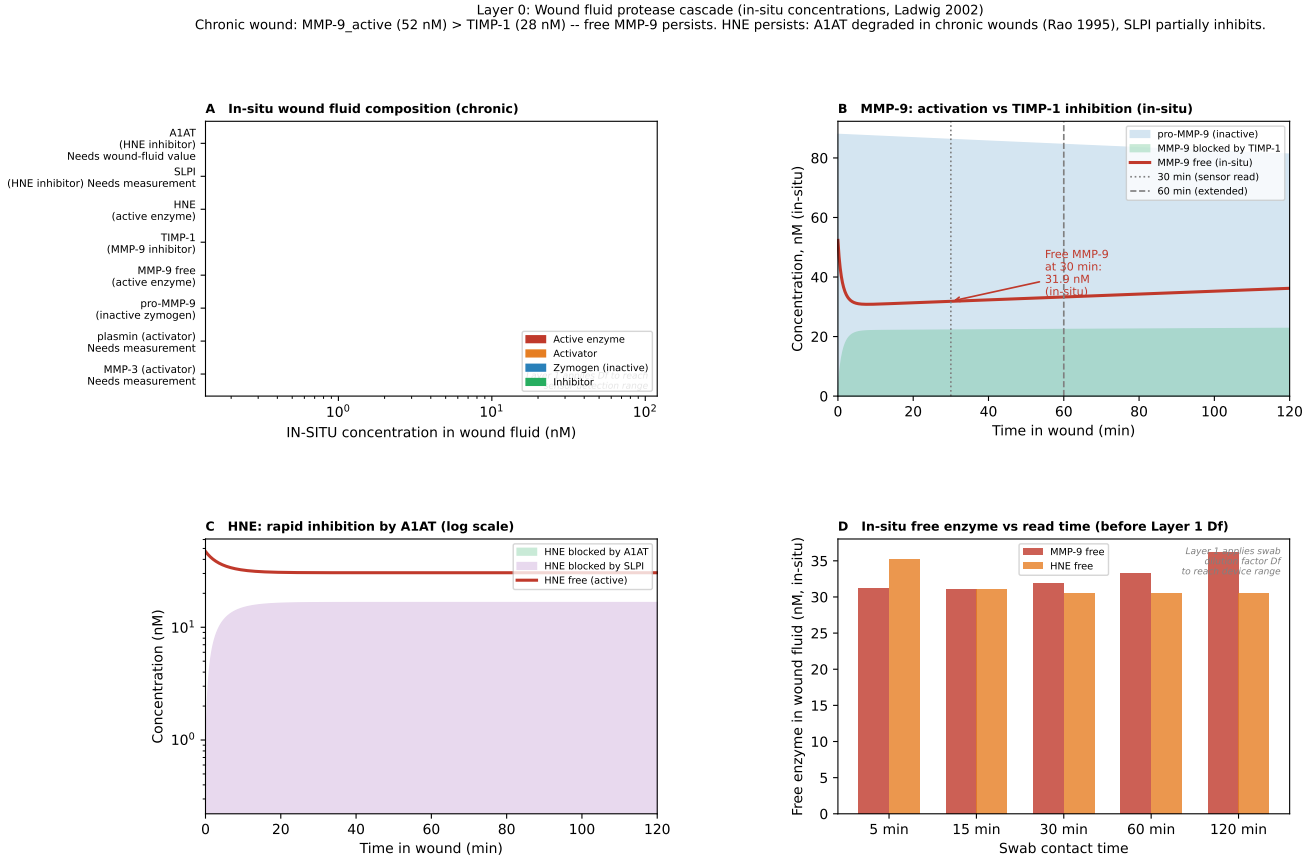


Figure 1: Layer 0 ODE output for the chronic wound state over 120 min. Top left: in-situ initial concentrations. Top right: MMP-9 activation vs. TIMP-1 inhibition. Bottom left: HNE in both wound states (log scale; note 5.5 ms half-life in healing). Bottom right: free enzyme at read times.

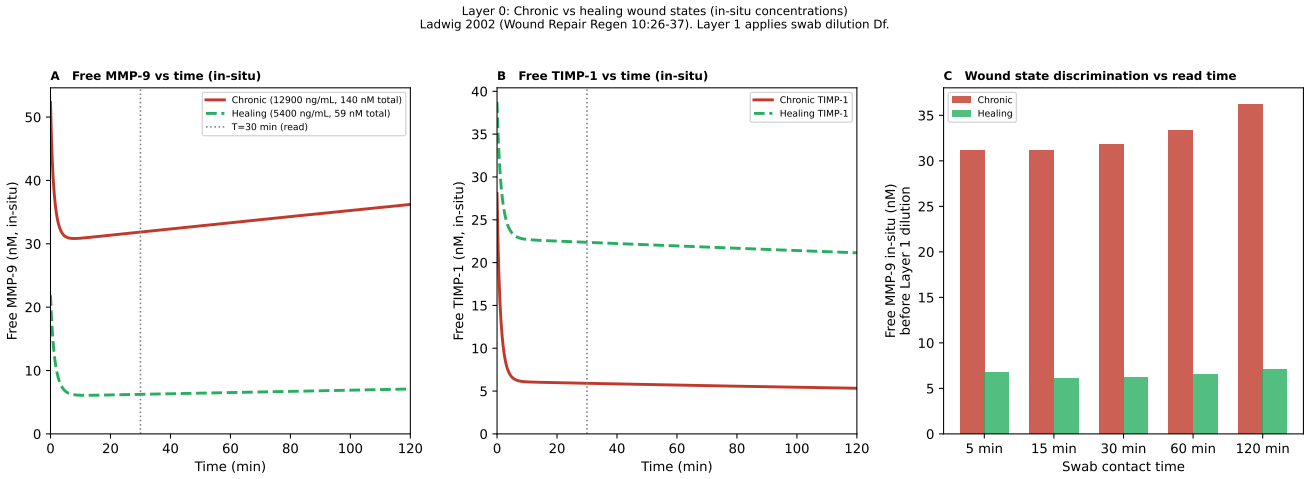


Figure 2: Chronic vs. healing wound states. Left: free MMP-9 over time. Centre: free TIMP-1. Right: MMP-9 discrimination at read times.

ODE results at $t = 30$ min

Species	Chronic wound	Healing wound	C/H ratio
Free MMP-9	31.9 nM	6.2 nM	5.1×
Free TIMP-1	5.9 nM	22.4 nM	–
Free HNE	30.5 nM †	0.00 pM	–
Free A1AT	0.00 μM	1.88 μM	–

Table 2: ODE output at $t = 30$ min. †A1AT degraded in 9/10 chronic samples (Rao 1995 [7]); HNE elevated in only 50% of chronic VLU (Trengove 1999 [1]). Chronic HNE prediction is sensitive to placeholder SLPI kinetics. Healing values depend on unconfirmed initial conditions.

MMP-9 wound-state discrimination

The 5.1× discrimination ratio arises from the sign of the MMP-9 vs. TIMP-1 excess, as made explicit by Eq. (II.9):

- **Chronic wound**: MMP-9 (52 nM) exceeds TIMP-1 (28 nM) by 24 nM. With $K_i = 8.5$ nM $\ll M_0, T_0$, TIMP-1 is nearly fully bound and free MMP-9 persists at ≈ 32 nM.
- **Healing wound** (estimated): TIMP-1 (39 nM) exceeds MMP-9 (22 nM) by 17 nM. TIMP-1 excess suppresses free MMP-9 to ≈ 6 nM.

The mechanism is structurally robust: for $K_i \ll M_0, T_0$, Eq. (II.9) gives $[M]_{\text{free}} \approx M_0 - T_0$ to first order (the excess simply remains free). Any wound state where active MMP-9 exceeds TIMP-1 will show elevated free MMP-9, independent of exact parameter values.

HNE wound-state discrimination

The A1AT wound-state switch (Rao 1995 [7], Section III) produces a strong binary in HNE: healing wounds have $[\text{HNE}]_{\text{free}} = 0.00$ pM, while chronic wounds with degraded A1AT show $[\text{HNE}]_{\text{free}} \approx 30$ nM. However, elevated HNE was observed in only 6/13 (50%) of chronic VLU samples (Trengove 1999 [1]); the remaining 50% of chronic wounds are HNE-low regardless of A1AT status. HNE is not a reliable primary discriminant; MMP-9 (5× discrimination, grounded in confirmed kinetics) is the preferred target. The second electrode targeting HNE serves as an internal reference rather than a primary channel.

Layer 1 dilution and sensor range

The sensor detection range (17–200 ng/mL MMP-9 = 0.185–2.17 nM) is reached only after Layer 1 dilution. The required dilution factor D_f is:

$$D_f = \frac{[\text{MMP-9}]_{\text{free, in-situ}}}{[\text{MMP-9}]_{\text{device}}} = \frac{32 \text{ nM}}{0.185\text{–}2.17 \text{ nM}} \approx 15\text{–}173 \tag{V.1}$$

With $D_f = 50$: chronic device entry = 32/50 = 0.64 nM = 59 ng/mL (within range); healing device entry = 6.2/50 = 0.12 nM = 11 ng/mL (near the lower LOD). The healing signal approaching the LOD under large D_f is a risk to quantify in Layer 1.

Limitations of this model

1. **Healing wound initial conditions not text-confirmed.** The healing MMP-9 estimate (5,400 ng/mL) is read from Ladwig 2002 Figure 8; the healing TIMP-1 (1,104 ng/mL) has no paper source. The ∼5× discrimination ratio should be treated as a working estimate until a joint VLU ELISA (MMP-9 and TIMP-1 in the same cohort at chronic and healing time points) is available.
2. **SLPI kinetics are unconfirmed placeholders.** The chronic wound HNE prediction (≈ 30 nM) is sensitive to the unconfirmed SLPI concentration and binding kinetics. Wound-fluid SLPI ELISA would substantially reduce this uncertainty.
3. **Pro-MMP-9 activation is negligible at current parameter values.** The activation kinetics (via MMP-3 and plasmin) use order-of-magnitude placeholders; $< 1\%$ of pro-MMP-9 activates in 120 min. These parameters become important only if wound-fluid MMP-3 or plasmin concentrations prove substantially higher than assumed.
4. **HNE elevation is inconsistent across chronic wound samples.** With elevated HNE in only 50% of chronic VLU cases, the HNE channel has limited diagnostic utility regardless of the kinetic model accuracy.
5. **Substrate competition with TIMP-1 and A1AT is not modelled.** The ODE treats inhibitor binding as an isolated equilibrium, without Michaelis–Menten substrate terms. In a full competitive system, the apparent inhibition constant would become $K_{i,\mathrm{app}} = K_i(1 + [S]/K_m)$. This simplification is justified here because (a) the sensor detects free MMP-9 *protein* via aptamer binding, not enzymatic activity, so the relevant quantity is the partition between free and TIMP-1-bound MMP-9, not substrate turnover; (b) TIMP-1 is a slow-tight inhibitor whose complex half-life ($t_{1/2} \approx 5.5$ min) greatly exceeds substrate catalytic cycle times (ms), so TIMP-1 wins the competition thermodynamically; and (c) the $K_i = 8.5$ nM from Olson 1997 [4] was measured in enzyme inhibition assays conducted in the presence of a fluorogenic substrate, so some substrate competition is already implicit in the reported value. For HNE, the serpin trapping mechanism means A1AT is not a classical competitive inhibitor, and the half-life conclusion is insensitive to a 10-fold change in k^A_{on}. If future layers require modelling MMP-9 *activity* rather than concentration, substrate terms would need to be added with wound-fluid substrate concentrations currently unavailable in any paper in the folder.

References

All papers in `electrochemical_device/papers/layer_00/`.

[1] Trengove NJ, Stacey MC, Macauley S, Bennett N, Gibson J, Burslem F, Murphy G, Schultz G (1999). Analysis of the acute and chronic wound environments: the role of proteases and their inhibitors. *Wound Repair Regen* 7(6):442–452. **trengove1999.pdf** – TIMP-1 and HNE values from text (p. 448) confirmed.

[2] Ladwig GP, Robson MC, Liu R, Kuhn MA, Muir DF, Schultz GS (2002). Ratios of activated matrix metalloproteinase-9 to tissue inhibitor of metalloproteinase-1 in wound fluids are inversely correlated with healing of pressure ulcers. *Wound Repair Regen* 10(1):26–37. **ladwig2002.pdf** – chronic MMP-9 and TIMP-1 from text (p. 31) confirmed; healing values from Figure 8 only (approximate).

[3] Nagase H, Woessner JF Jr (1999). Matrix metalloproteinases. *J Biol Chem* 274(31):21491–21494. **nagase_woessner1999.pdf** – biological context; no kinetic constants used.

[4] Olson MW, Gervasi DC, Mobashery S, Fridman R (1997). Kinetic analysis of the binding of human matrix metalloproteinase-2 and -9 to tissue inhibitor of metalloproteinase (TIMP)-1 and TIMP-2. *J Biol Chem* 272(47):29975–29983. **olson1997.pdf** – k^T_{on}, k^T_{off}, K_i for MMP-9/TIMP-1 confirmed from Table IV.

[5] Beatty K, Bieth J, Travis J (1980). Kinetics of association of serine proteinases with native and oxidized α-1-proteinase inhibitor and α-1-antichymotrypsin. *J Biol Chem* 255(9):3931–3934. **beatty1980.pdf** – $k^A_{\mathrm{on}} = 6.5 \times 10^7\ \mathrm{M^{-1}s^{-1}}$ confirmed from Table 1.

[6] Ashcroft GS, Lei K, Jin W, Longenecker G, Kulkarni AB, Greenwell-Wild T, Hale-Donze H, McGrady G, Song XY, Wahl SM (2000). Secretory leukocyte protease inhibitor mediates non-redundant functions necessary for normal wound healing. *Nat Med* 6(10):1147–1153. **ashcroft2000.pdf** – SLPI role confirmed in knockout mouse model; no quantitative wound-fluid SLPI concentration reported.

[7] Rao CN, Ladin DA, Liu YY, Chilukuri K, Dong Z, Bhagwat SV (1995). α_1-Antitrypsin is degraded and non-functional in chronic wounds but intact and functional in acute wounds: the inhibitor protects fibronectin from degradation by chronic wound fluid enzymes. *J Invest Dermatol* 105(4):572–578. **rao1995.pdf** – A1AT degraded in 9/10 chronic samples, intact in healing, confirmed from text (pp. 572–575). No quantitative wound-fluid A1AT concentration given.

[8] Nwomeh BC, Liang HX, Cohen IK, Yager DR (1999). MMP-8 is the predominant collagenase in healing wounds and nonhealing ulcers. *J Surg Res* 81(2):189–195. **nwomeh1999.pdf** – TIMP-1 "very low or not detected" in

chronic ulcer fluid confirmed from text (p. 193). Units not convertible to ng/mL without protein content.